

[Iodine: Fact Sheet for Consumers \(original version\)](#)

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What is iodine and what does it do?

Iodine is a mineral found in some foods. The body needs iodine to make thyroid hormones. These hormones control the body's metabolism and many other important functions. The body also needs thyroid hormones for proper bone and brain development during pregnancy and infancy. Getting enough iodine is important for everyone, especially infants and women who are pregnant.

How much iodine do I need?

The amount of iodine you need each day depends on your age. Average daily recommended amounts are listed below in micrograms (mcg).

Life Stage	Recommended Amount
Birth to 6 months	110 mcg
Infants 7–12 months	130 mcg
Children 1–8 years	90 mcg
Children 9–13 years	120 mcg
Teens 14–18 years	150 mcg
Adults	150 mcg
Pregnant teens and women	220 mcg
Breastfeeding teens and women	290 mcg

What foods provide iodine?

Iodine is found naturally in some foods and is also added to salt that is labeled as iodized. You can get recommended amounts of iodine by eating a variety of foods, including the following:

- Fish (such as cod and tuna), seaweed, shrimp, and other seafood, which are generally rich in iodine
- Dairy products (such as milk, yogurt, and cheese) and eggs, which are also good sources of iodine

- Iodized salt, which is readily available in the United States and many other countries*

*Processed foods, such as canned soups, almost never contain iodized salt. In addition, specialty salts, such as sea salt, kosher salt, Himalayan salt, and fleur de sel, are not usually iodized. Product labels will indicate if the salt is iodized or provides iodide.

What kinds of iodine dietary supplements are available?

Iodine is available in dietary supplements, usually in the form of potassium iodide or sodium iodide. Many multivitamin/mineral supplements contain iodine. Dietary supplements of iodine-containing kelp (a seaweed) are also available.

Am I getting enough iodine?

Most people in the United States get enough iodine from foods and beverages. However, certain groups of people are more likely than others to have trouble getting enough iodine:

- People who do not use iodized salt. Adding iodine to salt is the most widely used strategy to control iodine deficiency. Currently, about 88% of households worldwide use iodized salt.
- Pregnant women. Women who are pregnant need about 50% more iodine than other women to provide enough iodine for their baby. Surveys show that many pregnant women in the United States may not get quite enough iodine, although experts do not know whether this affects their babies.
- People who follow a vegan diet or who eat few or no dairy products, seafood, and eggs. Seafood, eggs, milk, and milk products are among the best sources of iodine. People who don't eat much of these foods or don't eat them at all might not get enough iodine.
- People living in regions with iodine-deficient soils who eat mostly local foods. These soils produce crops that have low iodine levels. Among the regions with the most iodine-poor soil are mountainous areas, such as the Himalayas, the Alps, and the Andes regions, as well as river valleys in South and Southeast Asia.
- People who get marginal amounts of iodine and who also eat foods containing goitrogens. Goitrogens are substances that interfere with the way the body uses iodine. They are present in some plant foods including soy and cruciferous vegetables such as cabbage, broccoli, cauliflower and brussels sprouts. For most people in the United States who get adequate amounts of iodine, eating reasonable amounts of foods containing goitrogens is not a concern.

What happens if I don't get enough iodine?

Iodine deficiency is uncommon in the United States and Canada. People who don't get enough iodine cannot make sufficient amounts of thyroid hormone. This can cause many problems. In pregnant women, severe iodine deficiency can permanently harm the fetus by causing stunted growth, intellectual disability, and delayed sexual development. Less severe iodine deficiency can cause lower-than-average IQ in infants and children and decrease adults' ability to work and think clearly. Goiter, an enlarged thyroid gland, is often the first visible sign of iodine deficiency.

What are some effects of iodine on health?

Scientists are studying iodine to understand how it affects health. Here are some examples of what this research has shown.

Fetal and infant development

Women who are pregnant or breastfeeding need to get enough iodine for their babies to grow and develop properly. Breastfed infants get iodine from breast milk. However, the iodine content of breast milk depends on how much iodine the mother gets.

To make adequate amounts of iodine available for proper fetal and infant development, several national and international groups recommend that pregnant and breastfeeding women and infants take iodine supplements. The American Thyroid Association recommends that women who are pregnant, planning to become pregnant, or breastfeeding take a daily supplement containing 150 mcg iodine as potassium iodide. The American Academy of Pediatrics has similar guidance. However, only about half the prenatal multivitamins sold in the United States contain iodine.

Cognitive function during childhood

Severe iodine deficiency during childhood has harmful effects on the development of the brain and nervous system. The effects of mild iodine deficiency during childhood are more difficult to measure, but mild iodine deficiency might cause subtle problems with neurological development.

Giving iodine supplements to children with mild iodine deficiency improves their reasoning abilities and overall cognitive function. In children living in iodine-deficient areas, iodine supplements seem to improve both physical and mental development. More study is needed to fully understand the effects of mild iodine deficiency and of iodine supplements on cognitive function.

Fibrocystic breast disease

Although not harmful, fibrocystic breast disease causes lumpy, painful breasts. It mainly affects women of reproductive age but can also occur during menopause. Very high doses of iodine supplements might reduce the pain and other symptoms of fibrocystic breast disease, but more study is necessary to confirm this. Check with your health care provider before taking iodine for this condition, especially because iodine can be unsafe at high doses.

Radiation-induced thyroid cancer

Nuclear accidents can release radioactive iodine into the environment, increasing the risk of thyroid cancer in people who are exposed to the radioactive iodine, especially children. People with iodine deficiency who are exposed to radioactive iodine are especially at risk of developing thyroid cancer. The U.S. Food and Drug Administration has approved potassium iodide as a thyroid-blocking agent to reduce the risk of thyroid cancer in radiation emergencies.

Low iodine diets for radioactive iodine treatment

Radioactive iodine treatment is used to treat some patients with thyroid cancer. Patients preparing for this treatment are usually told to follow a low iodine diet (50 mcg or less per day) for 1 to 2 weeks before treatment begins.

Foods that should be avoided when following a low iodine diet include:*

- Iodized salt
- Fish and other seafood, liver, seaweed, milk and milk products, and eggs
- Bread that is made with iodate dough conditioners (check the ingredient listings on product labels)
- Maraschino cherries and some red or pink beverages because these products are often made with iodine-containing red dyes
- Some dietary supplements such as kelp products and many multivitamin/mineral supplements (check the Supplement Facts label to see if iodine is listed). You can find products and view their Supplement Facts labels by searching the [Dietary Supplement Label Database](https://ods.od.nih.gov/api/?resourcename=Iodine&readinglevel=Consumer&outputformat=HTML).

* Nutrition Facts labels on foods usually list iodine only when the manufacturer has added iodine to the food. Foods that naturally contain iodine, such as seaweed and fish, will not usually show iodine on Nutrition Facts labels. Therefore, these labels cannot be relied upon to identify foods that contain iodine.

Foods that are lower in iodine and therefore okay to consume when following a low iodine diet include:

- Salt that is not iodized**
- Fruits and fruit juices, vegetables, beans, and nuts
- Plant-based milk substitutes (such as soy and almond beverages)
- Rice, pasta, and oatmeal
- Poultry, pork, and beef (except liver)
- Bread that is not made with iodate dough conditioners (check the ingredient listings on product labels)

** Salt that is not iodized (including many sea salts) is available in grocery stores. In addition, almost all salt used in processed foods in the United States is not iodized; product labels will indicate if the manufacturer used salt that is iodized or provides iodine.

The following resources may be helpful when planning low iodine diets:

- “[Low Iodine Diet](#)” from the American Thyroid Association provides guidelines and menu options for meals and snacks.
- The [USDA, FDA, and ODS-NIH Database for the Iodine Content of Common Foods](#) lists the iodine content of hundreds of foods and beverages.

Can iodine be harmful?

Yes, if you get too much. Getting high levels of iodine can cause some of the same symptoms as iodine deficiency, including goiter (an enlarged thyroid gland). High iodine intakes can also cause thyroid gland inflammation and thyroid cancer. Getting a very large dose of iodine (several grams, for example) can cause burning of the mouth, throat, and stomach; fever; stomach pain; nausea; vomiting; diarrhea; weak pulse; and coma.

The daily upper limits for iodine include intakes from all sources—food, beverages, and supplements—and are listed below. These levels do not apply to people who are taking iodine for medical reasons under the care of a doctor.

Life Stage	Upper Limit
Birth to 12 months:	Not established
Children 1–3 years:	200 mcg
Children 4–8 years:	300 mcg
Children 9–13 years:	600 mcg
Teens 14–18 years:	900 mcg
Adults:	1,100 mcg

Does iodine interact with medications or other dietary supplements?

Yes. Iodine supplements can interact or interfere with medicines that you take. Here are several examples:

- Iodine supplements might interact with antithyroid medications such as methimazole (Tapazole), used to treat hyperthyroidism. Taking high doses of iodine with antithyroid medications could cause your body to produce too little thyroid hormone.
- Taking potassium iodide with medicines for high blood pressure known as ACE inhibitors could raise the amount of potassium in your blood to an unsafe level. ACE inhibitors include benazepril (Lotensin), lisinopril (Prinivil and Zestril), and fosinopril (Monopril).
- The amount of potassium in your blood can also get too high if you take potassium iodide with potassium-sparing diuretics, such as spironolactone (Aldactone) and amiloride (Midamor).

Tell your doctor, pharmacist, and other health care providers about any dietary supplements and prescription or over-the-counter medicines you take. They can tell you if these dietary supplements might interact with

your medicines. They can also explain whether the medicines you take might interfere with how your body absorbs or uses iodine or other nutrients.

Iodine and healthful eating

People should get most of their nutrients from food and beverages, according to the federal government's *Dietary Guidelines for Americans*. Foods contain vitamins, minerals, dietary fiber, and other components that benefit health. In some cases, fortified foods and dietary supplements are useful when it is not possible to meet needs for one or more nutrients (for example, during specific life stages such as pregnancy). For more information about building a healthy dietary pattern, see the [Dietary Guidelines for Americans](#).

Where can I find out more about iodine?

- For general information on iodine
 - [Office of Dietary Supplements Health \(ODS\) Professional Fact Sheet on Iodine](#)
 - [Iodine in diet](#), MedlinePlus
- For more advice on choosing dietary supplements
 - [ODS Frequently Asked Questions: Which brand\(s\) of dietary supplements should I purchase?](#)
- For information about building a healthy dietary pattern
 - [Dietary Guidelines for Americans](#)

Glossary

absorption

In nutrition, the process of moving protein, carbohydrates, fats, and other nutrients from the digestive system into the bloodstream. Most absorption occurs in the small intestine.

ACE inhibitor

Angiotensin-converting enzyme inhibitor. A medicine used to treat high blood pressure, stroke, heart attack, diabetes, and kidney problems

agent

In medicine, a drug, dietary supplement, other substance, or procedure that is used in diagnosing, screening, preventing, or treating a disease.

cancer

A group of diseases in which cells divide abnormally and without control, and spread to nearby tissues and other parts of the body. Without treatment, cancer can stop organs from working normally, damage body systems, and cause the patient to die. Cancer may be caused by multiple factors, such as radiation, sunlight, tobacco, certain viruses, and poisonous chemicals; however, the cause of many cancers is unknown.

cognitive function

Mental awareness and judgment.

cruciferous vegetable

A type of vegetable including arugula, bok choy, broccoli, Brussels sprouts, cabbage, cauliflower, collards, kale, kohlrabi, mustard greens, radishes, rutabaga, turnips and watercress.

deficiency

An amount that is not enough; a shortage.

diarrhea

Loose, watery stools.

dietary fiber

A substance in plants that you cannot digest. It adds bulk to your diet to make you feel full, helps prevent constipation, and may help lower the risk of heart disease and diabetes. Good sources of dietary fiber include whole grains (such as brown rice, oats, quinoa, bulgur, and popcorn), legumes (such as black beans, garbanzo beans, split peas, and lentils), nuts, seeds, fruit, and vegetables.

dietary supplement

A product that is intended to supplement the diet. A dietary supplement contains one or more dietary ingredients (including vitamins, minerals, herbs or other botanicals, amino acids, and other substances)

or their components; is intended to be taken by mouth as a pill, capsule, tablet, or liquid; and is identified on the front label of the product as being a dietary supplement.

diuretic

A drug or other substance that increases the amount of urine made by the body.

dose

The amount of medicine or other substance taken at one time or over a specific period of time.

fetal

Having to do with a fetus (the developing human from 7 to 8 weeks after conception until birth).

fetus

The developing human from 7 to 8 weeks after conception until birth.

fibrocystic breast disease

A common condition in which the breasts feel bumpy, tender, and painful, especially before a menstrual period.

Food and Drug Administration

FDA, Department of Health and Human Services. FDA is the Federal government agency responsible for ensuring that foods and dietary supplements are safe, wholesome and sanitary, and that drugs, medical devices, cosmetics, and food are honestly, accurately and informatively represented to the public. FDA regulates dietary supplements under a different set of regulations than those covering conventional foods and drug products (prescription and over-the-counter). The dietary supplement manufacturer is responsible for ensuring that a dietary supplement is safe before it is marketed. FDA is responsible for taking action against any unsafe dietary supplement product after it reaches the market. Generally, manufacturers do not need to get FDA approval before producing or selling dietary supplements.

fortified

When nutrients (such as vitamins and minerals) are added to a food product. For example, when calcium is added to orange juice, the orange juice is said to be "fortified with calcium". Similarly, many breakfast cereals are "fortified" with several vitamins and minerals.

gland

A small organ that makes and releases a substance such as sweat, tears, saliva, milk, a hormone, or substances that aid in digestion.

goiter

An enlarged thyroid gland. A goiter is caused by too little iodine in the diet or by other conditions, such as a growth on the thyroid or a gland that makes too much or not enough hormones.

high blood pressure

A blood pressure measurement of 140/90 mmHg (millimeters of mercury) or higher is considered high blood pressure (hypertension). Blood pressure is the force of blood pushing against the walls of the arteries. Blood pressure measurements are written as two numbers, for example 120/80. The first number (the systolic pressure) measures the pressure when the heart beats and pumps out blood into the arteries. The second number (the diastolic pressure) measures the pressure when the heart is at rest between beats. High blood pressure is a condition that occurs when a person's blood pressure often measures above 140/90 or regularly stays at that level or higher. This condition usually has no symptoms but can be life-threatening. It damages the arteries and increases the chance of stroke, heart attack, kidney failure, and blindness. Also called hypertension.

hyperthyroidism

A condition in which your thyroid gland makes more hormone than your body needs. Symptoms include weight loss, fatigue, restlessness, frequent bowel movements or diarrhea, and goiter.

inflammation

Redness, swelling, pain, and/or a feeling of heat in an area of the body. It is a protective reaction to injury, disease, or irritation of tissues.

interaction

A change in the way a dietary supplement acts in the body when taken with certain other supplements, medicines, or foods, or when taken with certain medical conditions. Interactions may cause the dietary supplement to be more or less effective, or cause effects on the body that are not expected.

menopause

The time of life when a woman's menstrual periods stop. A woman is in menopause when she hasn't had a period for 12 months in a row. Also called "change of life."

mineral

In nutrition, an inorganic substance found in the earth that is required to maintain health.

nausea

The uneasy feeling of having an urge to throw up (vomit).

nervous system

The brain and spinal cord, including the network of nerves that carry messages back and forth between the brain and all parts of the body. The nervous system controls what the body does.

neurologic

Having to do with nerves and the nervous system.

nutrient

A chemical compound in food that is used by the body to function and maintain health. Examples of nutrients include proteins, fats, carbohydrates, vitamins, and minerals.

pharmacist

A person licensed to make and dispense (give out) prescription drugs and who has been taught how they work, how to use them, and their side effects.

prenatal

Before birth; during pregnancy.

prescription

A written order from a health care provider for medicine, therapy, or tests.

radioactive iodine

A type of iodine that is created by nuclear power plants and nuclear bombs. It can enter your body if you are exposed to contaminated air, food, or water. In the body, radioactive iodine collects in your thyroid and can damage the gland. At careful doses, radioactive iodine is also used in nuclear medicine. It is used in imaging tests to diagnose thyroid problems, and in cancer treatment to kill thyroid tumor cells.

reproductive age

The time of life during which a person is able to conceive a child. It begins in puberty (for men and women) and ends after menopause (for women only).

risk

The chance or probability that a harmful event will occur. In health, for example, the chance that someone will develop a disease or condition.

sign

An indication of disease that can be seen and/or measured. Examples include high fever, high blood pressure, infection, and coughing up blood.

soy

A plant that produces beans used in many food products. Soy products contain isoflavones (estrogen-like substances) that are being studied in the prevention of cancer, hot flashes that occur with menopause, and osteoporosis (loss of bone density). Also called soya and soybean. Latin name: *Glycine max*.

stunted growth

Not able to reach the expected height, size, or level of development for a child's age.

symptom

A feeling of sickness that an individual can sense, but that cannot be measured by a healthcare professional. Examples include headache, tiredness, stomach ache, depression, and pain.

treat

To care for a patient with a disease by using medicine, surgery, or other approaches.

vitamin

A nutrient that the body needs in small amounts to function and maintain health. Examples are vitamins A, C, and E.

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